Forecasting NYC Traffic Injuries Up to 30% Better

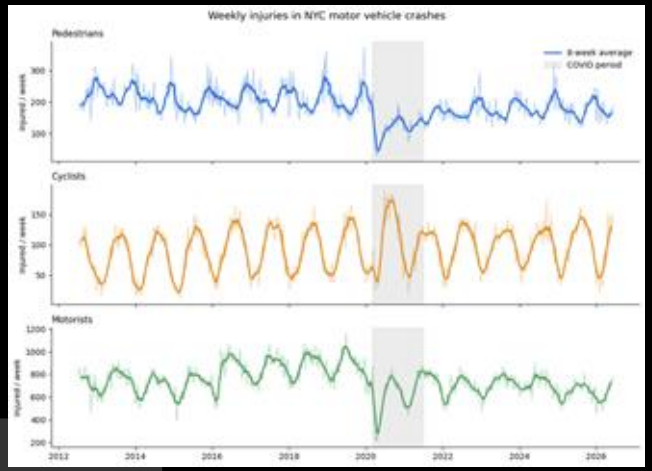
DS 4002 - Section 001

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Motivation → Data → Methods → Results → Implications

Motivation:
Why Forecast Traffic Injuries?



Traffic injuries not evenly distributed throughout the year

Our exploratory analysis found distinct seasonal patterns across the three road-user groups

Different road users experience different seasonal patterns

From our EDA: Pedestrian injuries peak in late fall and early winter, cyclist injuries peak during midsummer, motorist injuries peak during early summer

Major disruptions can change these patterns

The COVID-19 pandemic caused significant changes in weekly traffic injuries, local policies also majorly affect the number of people on the roads

Our Research Question

How many pedestrians, cyclists, and motorists will be injured in motor vehicle crashes in NYC over a held-out 12-week period, and what seasonal and trend patterns drive those counts?



Goal: We will process NYPD collision records to train and evaluate SARIMAX forecasting models using rolling-origin cross-validation and STL decomposition, achieving success if each model's MAE over the final 12-week test period of 2025 is at least 10% lower than a seasonal-naive baseline and at least 80% of observed counts fall within its 80% prediction intervals.

Data Acquisition & Explanation

Source: NYC Open Data

Ethics Note

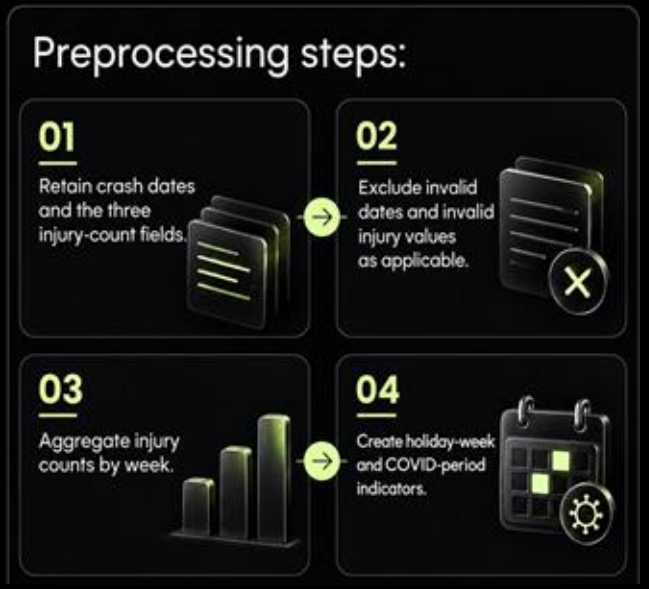
Records describe real people, so we only analyze weekly totals and never individual crashes

2.27M

Crash records – we used data from 2016-2025

29

Original features (crash date, location, number of motorists, pedestrians, cyclists injured / killed



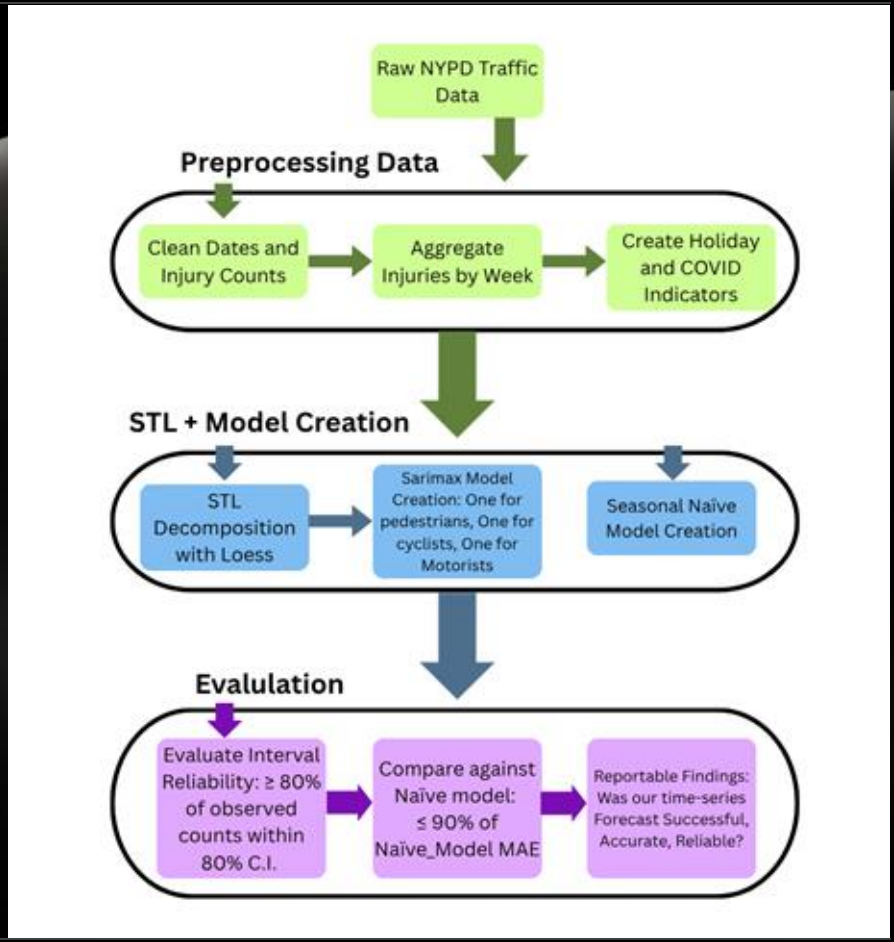
CSV format, one row for each collision, free to download under the NYC Open Data Terms of Use

Analysis Pipeline

Sritha – Data Preprocessing
Eileen – SARIMAX Model Development
Hayden – STL Decomposition + Evaluation

Validation Work

Rolling-origin cross-validation: 280-week initial training window, 18 folds of 12 weeks each; all 54 cross-validation fits converged



Tricky Analysis Decision

How should we improve forecast accuracy without overfitting?



The challenge: The initial SARIMAX configurations did not perform equally well across all three road-user groups. More complex models might improve accuracy, but could also overfit historical patterns.

The decision: Test alternative SARIMAX orders, guided by validation performance and residual diagnostics, rather than using identical configurations for all three groups.

This went beyond our MI2 plan, which only compared a predefined set of models before testing

The impact: Revised models reduced pedestrian test-period MAE by approximately 37% and cyclist test-period MAE by 36%, while motorist MAE remained unchanged

Bias + Uncertainty

Reporting Bias

The NYPD dataset includes police-reported collisions, meaning some crashes may not be represented.

Change in reporting practice

the NYPD transitioned to electronic collision reporting in March 2016

→ We start our data in March 2016 to avoid this

Uncertainty

Each forecast has an 80% prediction interval from SARIMAX. We checked these intervals on the cross-validation folds (91% / 84% / 98% coverage for pedestrians/cyclists/motorists) and on the test period (100% / 92% / 100%

COVID-19 disruptions

The pandemic significantly changed traffic patterns and injury counts

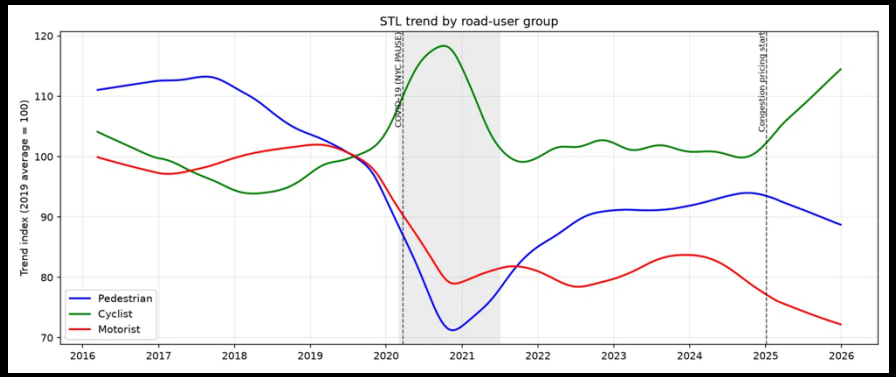
→ We used COVID-19 indicators to build our model

Unmeasured external factors

Changes in traffic volume, weather, and street infrastructure may influence injuries without being explicitly modeled.

Results





1. **Forecast accuracy:** Revised SARIMAX models achieved MAE reductions of 30.08%, 14.32%, and 17.17% for pedestrians, cyclists, and motorists, respectively, compared with the seasonal-naive baseline.
2. **Seasonality:** STL identified distinct seasonal patterns. Pedestrian injuries peaked around week 50, cyclist injuries around week 31, and motorist injuries around week 26.
3. **Uncertainty:** All three revised models exceeded the target of 80% empirical coverage during the examined 12-week period, although this is not independent validation.
4. **Goal Met:** All three models beat the baseline by more than 10% and exceeded 80% interval coverage
5. **Extra:** The STL results also show particularly strong seasonality for cyclists (seasonal strength 0.810), compared with pedestrians (0.447) and motorists (0.411).

Next Steps

1

Validate on new data

Evaluate the revised models on a genuinely untouched time period to test whether improvements generalize.

2

Improve model calibration

Investigate residual autocorrelation, particularly for cyclists, and assess prediction interval width alongside coverage.

3

Investigate external influences

Explore whether changes such as NYC's 2025 congestion pricing policy are associated with changes in injury trends, without assuming causation.

References

Thank you to Professor Alonzi and Naman for feedback and support, as well as NYC Open Data for the dataset.

[1] New York City Police Department, “Motor Vehicle Collisions – Crashes,” NYC Open Data. Accessed: Oct. 7, 2026. [Online]. Available: https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95

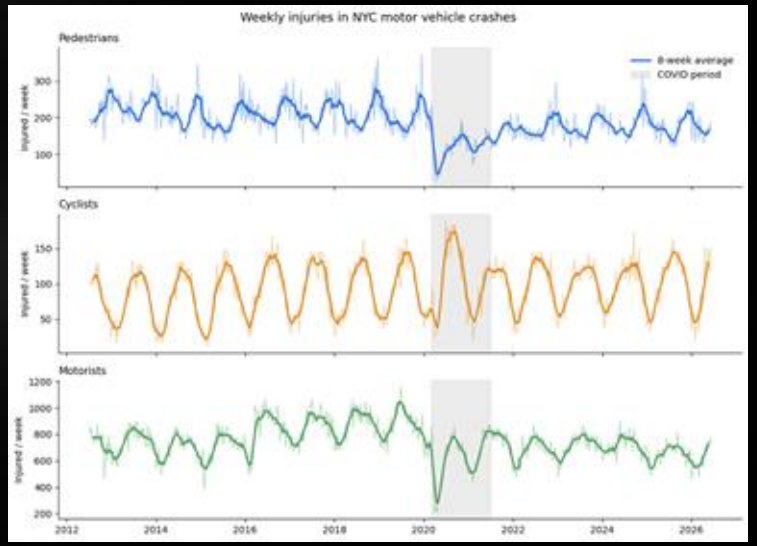
[2] NYC Open Data, “Terms of Use.” Accessed: Oct. 15, 2026. [Online]. Available: https://data.cityofnewyork.us/stories/s/Terms-of-Use/k9k7-3cje

[3] D. Gannon, “One year of congestion pricing in NYC, 27 million fewer vehicles, $550M in revenue,” 6sqft, Jan. 5, 2026. Accessed: Oct. 16, 2026. [Online]. Available: https://www.6sqft.com/one-year-of-congestion-pricing-in-nyc-27-million-fewer-vehicles-less-traffic/?p=230110

[4] Code and data: https://github.com/eileenkopp/DS4002-project2/tree/main

Thank you


Average yearly seasonal pattern by road-user group
Pedestrian
Cyclist
Motorist
Seasonal effect (% of mean trend)
Week of year


Weekly injuries in NYC motor vehicle crashes
Pedestrians
8-week average
COVID period
Injured / week
Cyclists
Motorists
2012
2014
2016
2018
2020
2022
2024
2026